

Exploratory Data Analysis - Music Dataset

C5056374

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1. Introduction

This report explores the music dataset provided by an online streaming company. The goal is to understand the data quality, find patterns, and help data analysts collect better data in the future.

2. Data Import and Setup

First, we will load the necessary libraries and the dataset.

```
# Load required libraries for analysis
library(tidyverse)  # For data manipulation and plotting
library(ggplot2)    # For graphs
library(dplyr)       # For data wrangling
library(corrplot)    # For correlation heatmaps
library(readr)       # For reading CSV files

# Loading the datasets
data <- read_csv("data.csv")

# Show first few rows
head(data)
```

```
## # A tibble: 6 x 19
##   valence  year acoustiness artists  danceability duration_ms energy explicit
##   <dbl> <dbl>      <dbl> <chr>          <dbl>      <dbl> <dbl> <dbl>
## 1  0.0594  1921      0.982 ['Sergei ~    0.279      831667  0.211      0
## 2  0.963   1921      0.732 ['Dennis ~    0.819      180533  0.341      0
## 3  0.0394  1921      0.961 ['KHP Kri~    0.328      500062  0.166      0
## 4  0.165   1921      0.967 ['Frank P~    0.275      210000  0.309      0
## 5  0.253   1921      0.957 ['Phil Re~    0.418      166693  0.193      0
## 6  0.196   1921      0.579 ['KHP Kri~    0.697      395076  0.346      0
## # i 11 more variables: id <chr>, instrumentalness <dbl>, key <dbl>,
## #   liveness <dbl>, loudness <dbl>, mode <dbl>, name <chr>, popularity <dbl>,
## #   release_date <chr>, speechiness <dbl>, tempo <dbl>
```

```
# Show dimensions(rows x columns)
cat("Dataset dimensions:", dim(data)[1], "rows and", dim(data)[2], "columns\n\n")
```

```
## Dataset dimensions: 170653 rows and 19 columns
```

```
# Show data types
str(data)
```

```
## spc_tbl_ [170,653 x 19] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ valence      : num [1:170653] 0.0594 0.963 0.0394 0.165 0.253 0.196 0.406 0.0731 0.721 0.771
## $ year         : num [1:170653] 1921 1921 1921 1921 1921 ...
## $ acousticness : num [1:170653] 0.982 0.732 0.961 0.967 0.957 0.579 0.996 0.993 0.996 0.982 ...
## $ artists      : chr [1:170653] "['Sergei Rachmaninoff', 'James Levine', 'Berliner Philharmoniker'
## $ danceability  : num [1:170653] 0.279 0.819 0.328 0.275 0.418 0.697 0.518 0.389 0.485 0.684 ...
## $ duration_ms   : num [1:170653] 831667 180533 500062 210000 166693 ...
## $ energy        : num [1:170653] 0.211 0.341 0.166 0.309 0.193 0.346 0.203 0.088 0.13 0.257 ...
## $ explicit      : num [1:170653] 0 0 0 0 0 0 0 0 0 0 ...
## $ id            : chr [1:170653] "4BJqT0PrAfrxzM0xytF0Iz" "7xPhfUan2yNtyFG0cUWkt8" "1o6I8Bg1A6yLD
## $ instrumentalness: num [1:170653] 8.78e-01 0.00 9.13e-01 2.77e-05 1.68e-06 1.68e-
01 0.00 5.27e-01 1.51e-01 0.00 ...
## $ key           : num [1:170653] 10 7 3 5 3 2 0 1 5 8 ...
## $ liveness      : num [1:170653] 0.665 0.16 0.101 0.381 0.229 0.13 0.115 0.363 0.104 0.504 ...
## $ loudness      : num [1:170653] -20.1 -12.44 -14.85 -9.32 -10.1 ...
## $ mode          : num [1:170653] 1 1 1 1 1 1 1 0 0 1 ...
## $ name          : chr [1:170653] "Piano Concerto No. 3 in D Minor, Op. 30: III. Finale. Alla brev
## $ popularity    : num [1:170653] 4 5 5 3 2 6 4 2 0 0 ...
## $ release_date   : chr [1:170653] "1921" "1921" "1921" "1921" ...
## $ speechiness    : num [1:170653] 0.0366 0.415 0.0339 0.0354 0.038 0.07 0.0615 0.0456 0.0483 0.399
## $ tempo         : num [1:170653] 81 60.9 110.3 100.1 101.7 ...
## - attr(*, "spec")=
## .. cols(
## ..   valence = col_double(),
## ..   year = col_double(),
## ..   acousticness = col_double(),
## ..   artists = col_character(),
## ..   danceability = col_double(),
## ..   duration_ms = col_double(),
## ..   energy = col_double(),
## ..   explicit = col_double(),
## ..   id = col_character(),
## ..   instrumentalness = col_double(),
## ..   key = col_double(),
## ..   liveness = col_double(),
## ..   loudness = col_double(),
## ..   mode = col_double(),
## ..   name = col_character(),
## ..   popularity = col_double(),
## ..   release_date = col_character(),
## ..   speechiness = col_double(),
## ..   tempo = col_double()
## .. )
## - attr(*, "problems")=<externalptr>
```

```
# Select only required columns
required_features <- c("acousticness", "artists", "danceability", "energy",
  "id", "liveness", "loudness", "name", "popularity",
  "speechiness", "tempo", "valence")
```

```
data_clean <- data %>% select(all_of(required_features))
```

```
# Check for the missing values
```

```
missing_values <- sum(is.na(data_clean))
```

```
cat("Total missing values in selected features:", missing_values, "\n")
```

```
## Total missing values in selected features: 0
```

```
# Remove the duplicates
```

```
data_clean <- data_clean %>% distinct()
```

```
cat("After removing duplicates:", nrow(data_clean), "rows remain\n")
```

```
## After removing duplicates: 170653 rows remain
```

```
# Check data types
```

```
sapply(data_clean, class)
```

```
## acousticness      artists danceability      energy      id      liveness
##      "numeric"    "character"    "numeric"    "numeric"    "character"    "numeric"
##      loudness      name      popularity    speechiness      tempo      valence
##      "numeric"    "character"    "numeric"    "numeric"    "numeric"    "numeric"
```

```
# Summary of numeric columns
```

```
summary(data_clean %>% select_if(is.numeric))
```

```
##      acousticness      danceability      energy      liveness
##      Min.      :0.0000      Min.      :0.0000      Min.      :0.0000      Min.      :0.0000
##      1st Qu.:0.1020      1st Qu.:0.4150      1st Qu.:0.2550      1st Qu.:0.0988
##      Median :0.5160      Median :0.5480      Median :0.4710      Median :0.1360
##      Mean   :0.5021      Mean   :0.5374      Mean   :0.4824      Mean   :0.2058
##      3rd Qu.:0.8930      3rd Qu.:0.6680      3rd Qu.:0.7030      3rd Qu.:0.2610
##      Max.   :0.9960      Max.   :0.9880      Max.   :1.0000      Max.   :1.0000
##      loudness      popularity      speechiness      tempo
##      Min.      : -60.000      Min.      : 0.00      Min.      :0.00000      Min.      : 0.00
##      1st Qu.: -14.615      1st Qu.: 11.00      1st Qu.:0.03490      1st Qu.: 93.42
##      Median : -10.580      Median : 33.00      Median :0.04500      Median :114.73
##      Mean   : -11.468      Mean   : 31.43      Mean   :0.09839      Mean   :116.86
##      3rd Qu.: -7.183      3rd Qu.: 48.00      3rd Qu.:0.07560      3rd Qu.:135.54
##      Max.    : 3.855      Max.   :100.00      Max.   :0.97000      Max.   :243.51
##      valence
##      Min.      :0.0000
##      1st Qu.:0.3170
##      Median :0.5400
##      Mean   :0.5286
##      3rd Qu.:0.7470
##      Max.    :1.0000
```

```
top_artists <- data_clean %>%
```

```
  count(artists) %>%
```

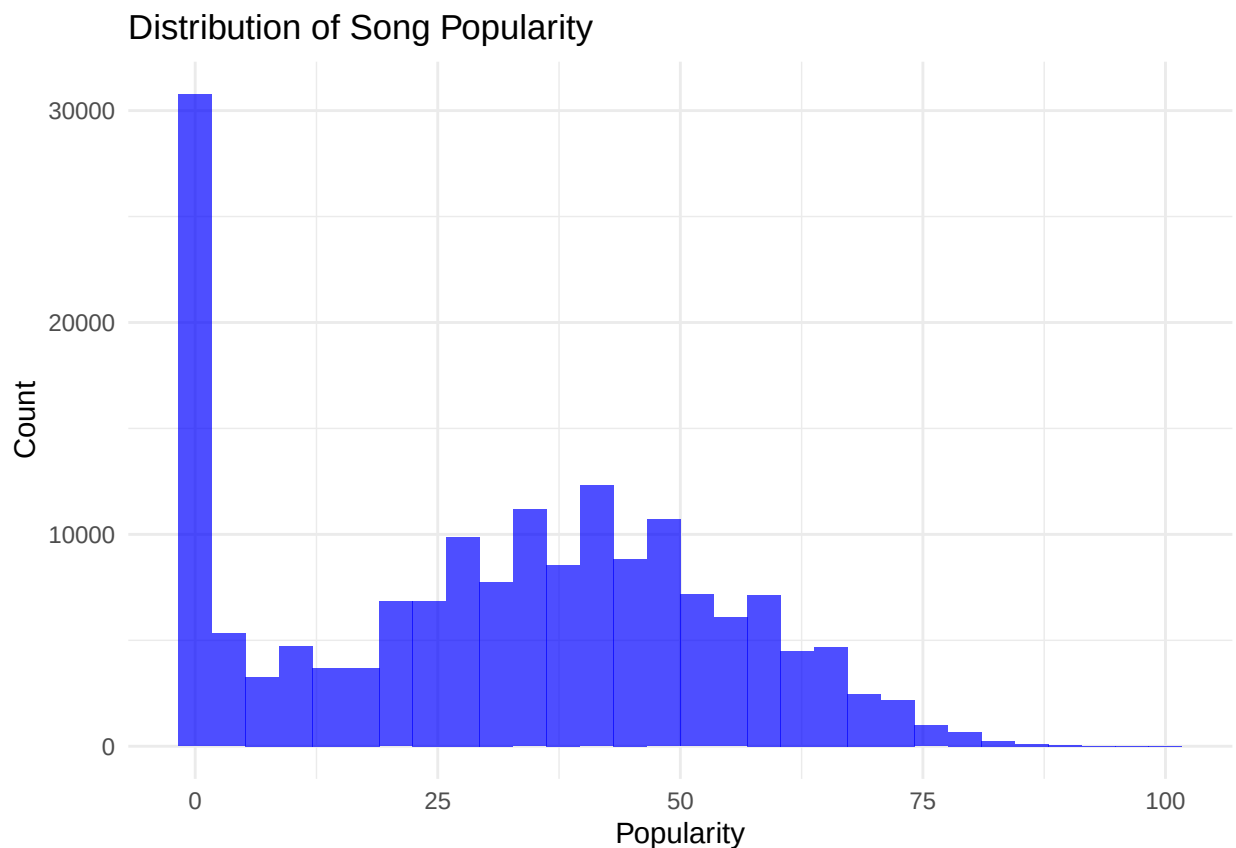
```
  arrange(desc(n)) %>%
```

```
head(10)
```

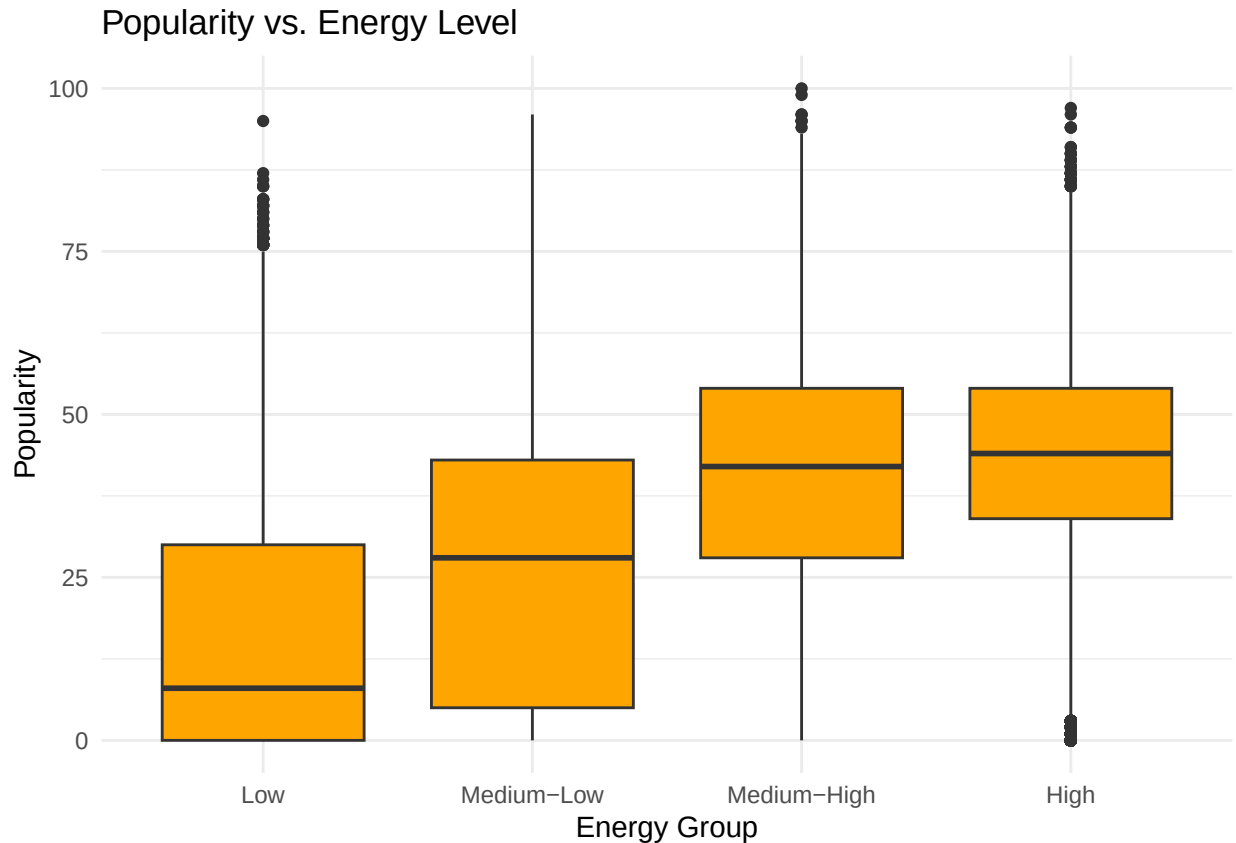
```
top_artists
```

```
## # A tibble: 10 x 2
##   artists          n
##   <chr>          <int>
## 1 ['          '] 1211
## 2 ['          '] 1068
## 3 ['Francisco Canaro'] 942
## 4 ['Frank Sinatra'] 630
## 5 ['Ignacio Corsini'] 628
## 6 ['Elvis Presley'] 504
## 7 ['Johnny Cash'] 484
## 8 ['Bob Dylan'] 474
## 9 ['The Rolling Stones'] 471
## 10 ['The Beach Boys'] 455
```

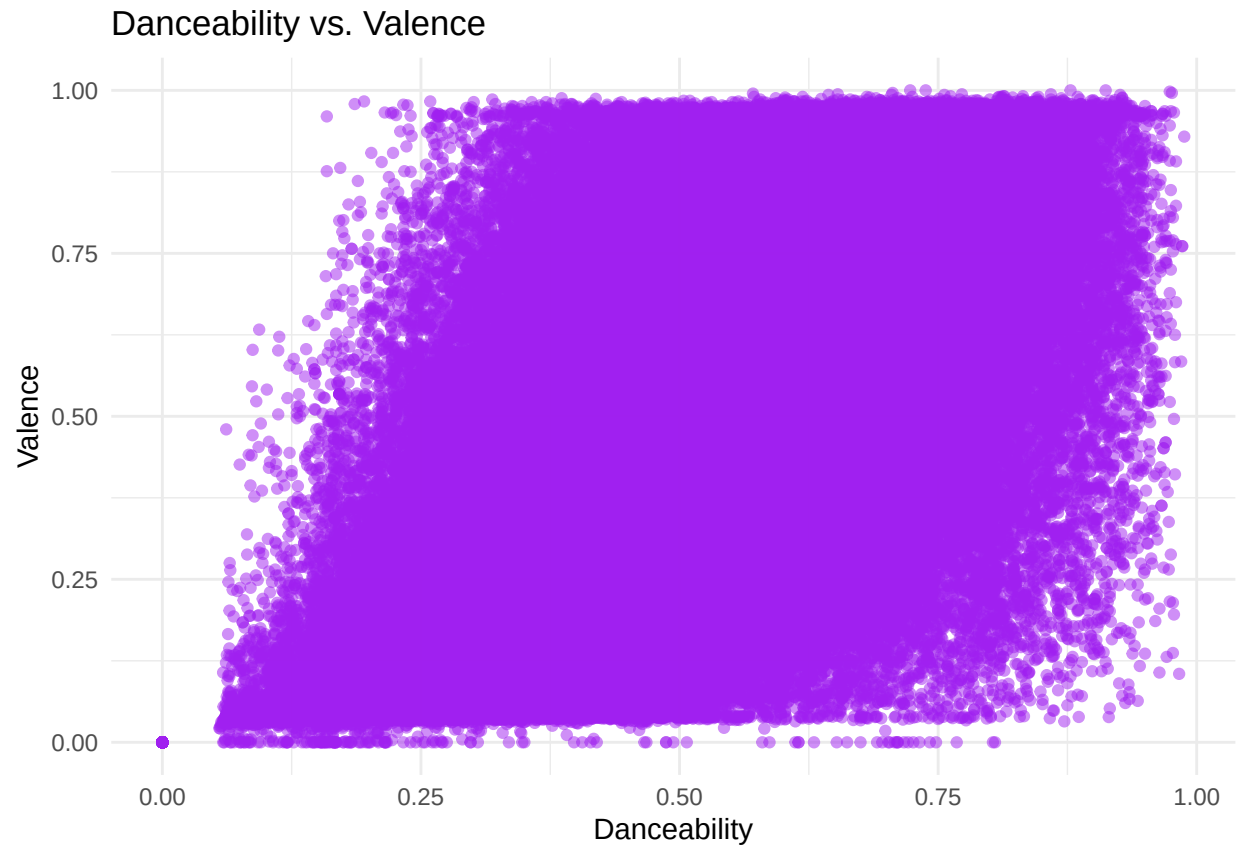
```
ggplot(data_clean, aes(x = popularity)) +
  geom_histogram(bins = 30, fill = "blue", alpha = 0.7) +
  labs(title = "Distribution of Song Popularity",
       x = "Popularity",
       y = "Count") +
  theme_minimal()
```



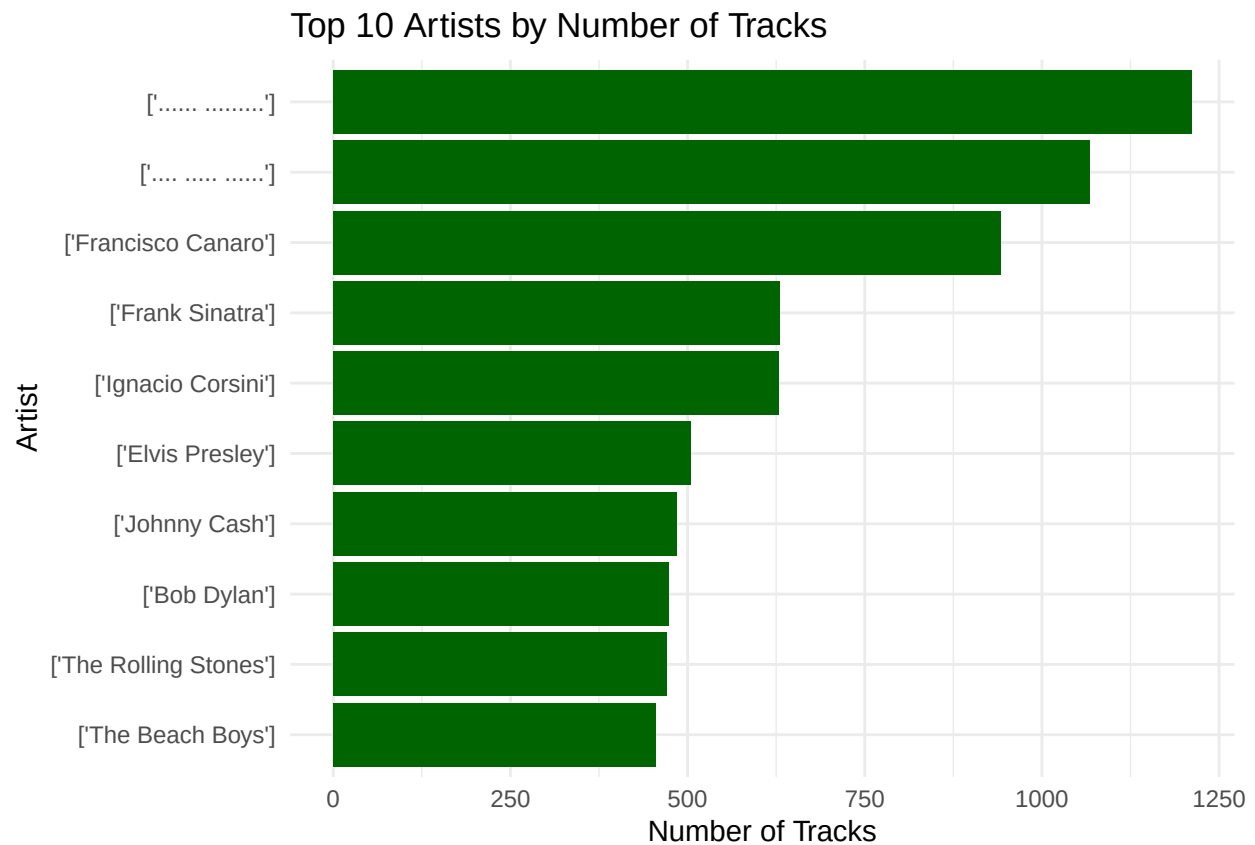
```
data_clean %>%
  mutate(energy_group = cut(energy, breaks = 4,
                            labels = c("Low", "Medium-Low", "Medium-High", "High"))) %>%
  ggplot(aes(x = energy_group, y = popularity)) +
  geom_boxplot(fill = "orange") +
  labs(title = "Popularity vs. Energy Level",
       x = "Energy Group",
       y = "Popularity") +
  theme_minimal()
```



```
ggplot(data_clean, aes(x = danceability, y = valence)) +
  geom_point(alpha = 0.5, color = "purple", size = 1.5) +
  labs(title = "Danceability vs. Valence",
       x = "Danceability",
       y = "Valence") +
  theme_minimal()
```



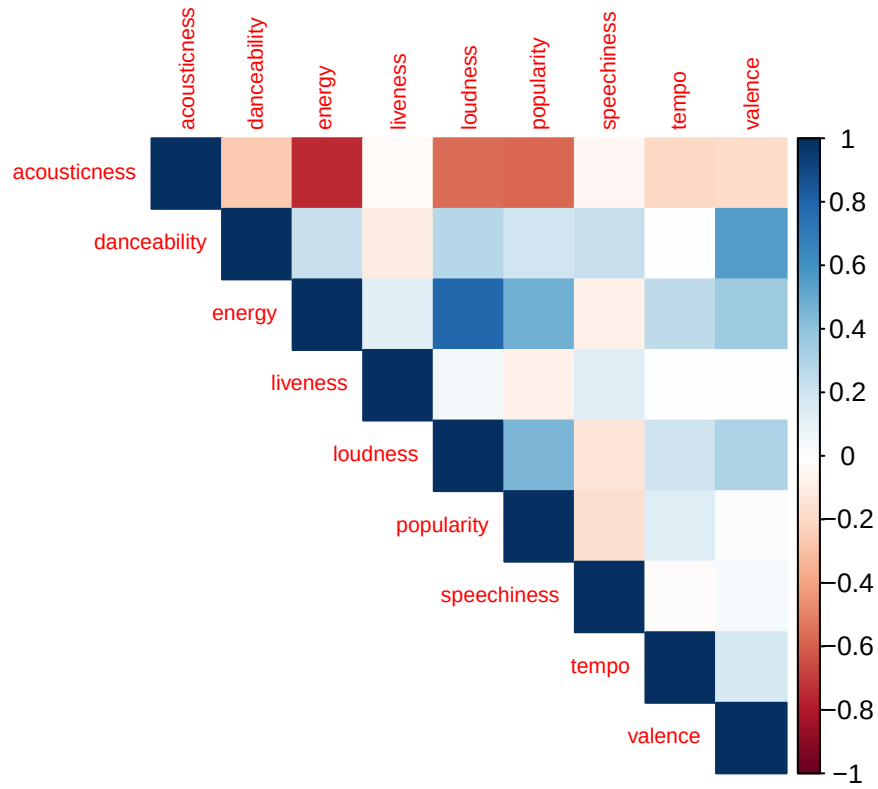
```
ggplot(top_artists, aes(x = reorder(artists, n), y = n)) +  
  geom_bar(stat = "identity", fill = "darkgreen") +  
  coord_flip() +  
  labs(title = "Top 10 Artists by Number of Tracks",  
        x = "Artist",  
        y = "Number of Tracks") +  
  theme_minimal()
```



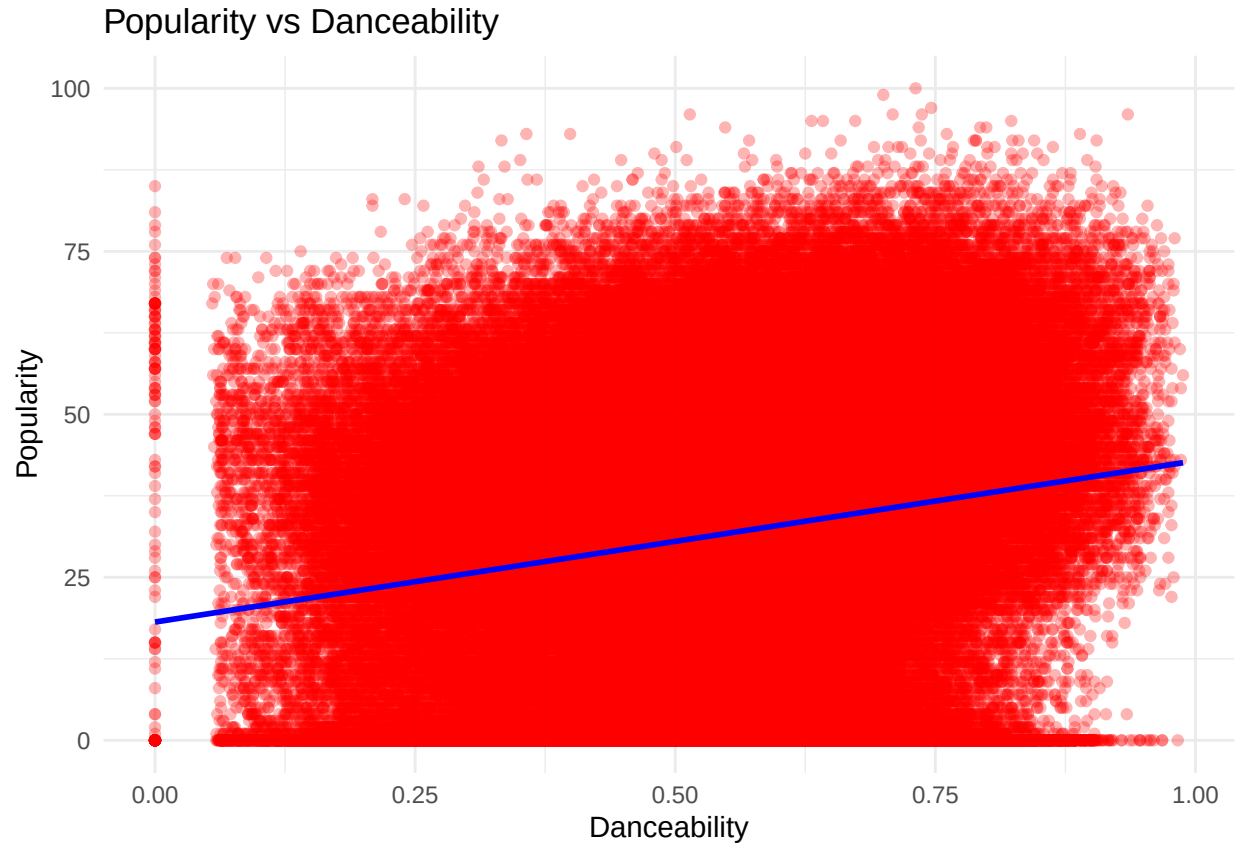
```
# Select only numeric columns for correlation
numeric_data <- data_clean %>% select_if(is.numeric)
cor_matrix <- cor(numeric_data, use = "complete.obs")

# Create correlation heatmap
corrplot(cor_matrix,
  method = "color",
  type = "upper",
  tl.cex = 0.7,
  title = "Correlation Between Music Features",
  mar = c(0, 0, 2, 0))
```

Correlation Between Music Features



```
# Relationship between popularity and danceability
ggplot(data_clean, aes(x = danceability, y = popularity)) +
  geom_point(alpha = 0.3, color = "red") +
  geom_smooth(method = "lm", color = "blue") +
  labs(title = "Popularity vs Danceability",
        x = "Danceability",
        y = "Popularity") +
  theme_minimal()
```



Appendix: AI Transparency Declaration Statement

AITs Descriptor: 1 – No AI

I confirm that no Artificial Intelligence (AI) tools were used in any part of this activity.

All aspects of this EDA report — including data import, cleaning, analysis, visualisations, interpretation, and writing — were human-generated, created, edited, and developed by me.

This statement is included as required by SHU's Academic Conduct Policy.